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Auditing the Faithfulness and Cross-City Consistency of Attention in Multimodal Spatiotemporal Air Quality Index Prediction

Results, Discussion and Inference

ML Lab Project - 20 October 2026 | Revised 2 November 2026

Including same-day estimation and next-day forecasting

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IV. Results, Discussion and Inference

A. Experimental Results

The completed experiments use distinct historical test periods and protocols. Within each comparison, models use the same eligible station-days. The 2024 same-day, 2025 next-day, 2026 paired forecast, 2023 chronological attention, and eight-city leave-one-city-out (LOCO) results should not be ranked across cohorts as though they shared one test distribution. The reserved October-December 2026 prospective period has no valid score yet.

A.1 Metric-wise performance and historical baselines

Table 1 shows that the 2024 same-day Ridge model with AQI history reduced RMSE from 36.197 for persistence to 32.806 AQI points on 28,803 station-days, a 9.4% reduction. Its MAE was 23.001, compared with 23.716 for persistence. Environment-only Ridge had RMSE 63.239. In the separate 2025 next-day test, the three-seed 28-day GRU reduced RMSE from 35.388 for persistence to 32.516 on 30,263 station-days, an 8.1% reduction. Its MAE was 21.857, compared with 22.981 for persistence.

Table 1 Historical performance against persistence. Each task has its own test cohort.

Model	Station-days	MAE	RMSE	R^2
Same-day estimation, 2024				
Ridge with AQI history	28,803	23.001	32.806	0.892
Persistence	28,803	23.716	36.197	0.869
Environment-only Ridge	28,803	45.804	63.239	0.599
Next-day forecast, 2025				
Three-seed GRU ensemble	30,263	21.857	32.516	0.887
Persistence	30,263	22.981	35.388	0.867

MAE and RMSE are in AQI points. Same-day estimation uses target-day environmental observations; next-day forecasting does not.

The paired weekly block-bootstrap RMSE gain was 3.391 AQI points [95% interval: 2.03, 4.67] for 2024 same-day Ridge and 2.872 [2.168, 3.605] for the 2025 GRU. GRU RMSE was lower than persistence in all eight cities in 2025. The nominal 90% station prediction interval covered 95.4% of the observed 2025 targets; this is historical coverage, not a future guarantee.

A.2 Paired 2026 forecast comparison

Weather-assisted XGBoost, the frozen GRU, and persistence were evaluated on exactly the same 20,086 station-days from January-September 2026 (91 of 107 configured stations). Table 2 and Fig. 1 show the metric-wise comparison. XGBoost had the lowest MAE and RMSE, and its RMSE was lower than GRU in each of the eight represented cities. Its paired RMSE gain was 1.437 points over GRU [95% weekly-bootstrap interval: 0.376, 2.614] and 4.398 points over persistence [3.190, 5.715].

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Table 2 Paired January-September 2026 forecast comparison.

Model	Station-days	MAE	RMSE	R^2
Weather-assisted XGBoost	20,086	19.345	30.270	0.848
Frozen GRU ensemble	20,086	20.394	31.707	0.833
Persistence	20,086	21.763	34.668	0.800

All three scores use identical rows. MAE and RMSE are in AQI points.

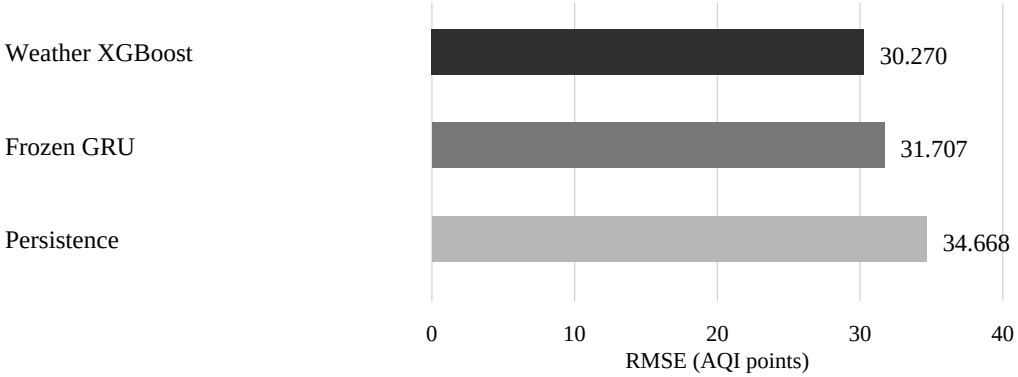


Fig. 1 RMSE on the same 20,086 January-September 2026 station-days. Lower is better; these bars do not compare different test years.

A.3 Abrupt-change and severe-AQI outcomes

Average error did not capture the full operational tradeoff. Of 2,315 observed station-days with an absolute daily AQI change of at least 50, XGBoost detected 422 changes in the correct direction, versus 154 for GRU. It flagged 799 changes overall, versus 268 for GRU (Table 3). XGBoost therefore had higher recall but lower precision.

Table 3 Sharp-change detection on the paired 2026 cohort; threshold = 50 AQI points.

Measure	Weather XGBoost	Frozen GRU
Observed sharp changes	2,315	2,315
Forecast-flagged changes	799	268
Correct directional detections	422	154
Recall	18.23%	6.65%
Precision	52.82%	57.46%

A correct detection must exceed the threshold in the observed direction. Precision is computed over forecast-flagged changes.

For observed AQI of 301-400, the weather model's mean prediction error was -38.51 points; for 401-500 it was -44.36 points. Only 94 observations were in the latter band, so that estimate is less stable. These severe-day errors matter even though the model led the average-error metrics.

A.4 Attention accuracy and faithfulness

The corrected attention study used a separate November-December 2023 chronological test of 5,434 station-days. Table 4 reports the SWT-w7 inputs (satellite, weather and day of year); Section A.6 shows that several outcomes change on SW-w7, which omits day of year. On SWT-w7, within the seven-day sequence family, temporal attention (TA), feature attention (FA), and their combination all had higher RMSE than uniform weighting. Within the spatial-patch family, spatial attention (S-SA) reduced RMSE from 75.328 to 65.836 (Table 4).

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Table 4 Corrected attention experiment, November-December 2023, SWT-w7 inputs.

Model family	Variant	Station-days	RMSE
Seven-day sequence	Uniform weights	5,434	78.361
Seven-day sequence	Temporal attention (TA)	5,434	84.394
Seven-day sequence	Feature attention (FA)	5,434	82.284
Seven-day sequence	TA + FA	5,434	94.525
Spatial patch	Uniform weights	5,434	75.328
Spatial patch	Temporal attention (S-TA)	5,434	83.641
Spatial patch	Spatial attention (S-SA)	5,434	65.836
Spatial patch	S-TA + S-SA	5,434	75.966

RMSE is in AQI points. Compare variants only within the same model family.

Faithfulness was audited on 300 sampled rows for each corrected SWT-w7 FA and S-SA checkpoint; no SW-w7 checkpoint has been audited. Table 5 reports row-weighted mean Spearman agreement between each ranking and fixed-model input-replacement sensitivity. SHAP exceeded attention under both replacement methods for both models; the size of the gap depended on the replacement method.

Table 5 Ranking agreement with input-replacement sensitivity.

Model	Replacement method	Attention r	SHAP r	Rows
FA	Mean replacement	0.420	0.664	300
FA	Same-city seasonal donor	0.392	0.546	300
S-SA	Mean replacement	0.242	0.419	300
S-SA	Same-city seasonal donor	0.483	0.586	300

Values are row-weighted mean Spearman correlations, not changes in forecast accuracy.

A.5 Corrected cross-city transfer

The completed SWT-w7 LOCO experiment trained each fold without one city and evaluated the held-out city on all eligible test rows. Across eight cities and 29,923 station-days, Table 6 reports both equal-city-weighted and pooled scores. Within the sequence family, TA had city-macro RMSE 55.23 versus 55.11 for Uniform, despite a slightly lower pooled RMSE (71.61 versus 71.67). Within the spatial-patch family, S-TA+SA reduced city-macro RMSE from 57.19 to 54.16 and pooled RMSE from 71.73 to 70.76. It improved five of eight held-out cities; sequence TA improved four of eight. These are separate historical transfer experiments, not prospective forecast scores.

Table 6 Corrected SWT-w7 LOCO evaluation: eight held-out cities and 29,923 station-days.

Family	Variant	City-macro MAE	City-macro RMSE	Pooled RMSE
Sequence	Uniform	40.42	55.11	71.67
Sequence	TA	40.45	55.23	71.61
Sequence	TA + FA	42.75	57.96	73.80
Sequence	FA	44.67	60.27	75.95
Spatial patch	S-Uniform	42.28	57.19	71.73
Spatial patch	S-TA	42.04	56.34	71.91
Spatial patch	S-SA	41.78	56.79	73.12
Spatial patch	S-TA + S-SA	40.20	54.16	70.76

MAE and RMSE are AQI points. City-macro averages give each held-out city equal weight; pooled RMSE weights station-days. Compare attention only with its within-family Uniform control.

A.6 Attention across input sets

The comparisons above use one input set. To test whether their conclusions depend on that choice, every attention variant was compared with its within-family Uniform control on both input sets and all three protocols, from the saved predictions and without retraining (Table 7). Some outcomes reversed. In the chronological test, S-SA gained 9.49 AQI points on SWT-w7 but lost 3.58 on SW-w7, and FA lost 3.92 on SWT-w7 but gained 3.59 on SW-w7. Sequence TA gained 1.73 in LOCO on SW-w7 but showed no gain on SWT-w7. The most consistent result was S-TA+SA in LOCO, which gained about 3 points on both input sets, although it showed no chronological gain on either.

Table 7 RMSE gain of each attention variant over its within-family Uniform control.

Family	Variant	SWT-w7 gain [95% interval]	SW-w7 gain [95% interval]
Chronological, November-December 2023			
Sequence	TA	-6.03 [-8.89, -2.49]	-0.75 [-3.57, 2.69]
Sequence	FA	-3.92 [-6.50, -0.38]	+3.59 [1.64, 5.33]
Sequence	TA + FA	-16.16 [-21.06, -9.83]	-7.27 [-11.96, -2.37]
Spatial patch	S-TA	-8.31 [-13.34, -3.80]	-1.31 [-4.88, 3.35]
Spatial patch	S-SA	+9.49 [3.64, 15.46]	-3.58 [-5.54, -1.61]
Spatial patch	S-TA + S-SA	-0.64 [-4.08, 2.25]	-2.68 [-6.76, 1.58]
Leave-one-city-out, eight cities			
Sequence	TA	-0.12 [-1.25, 0.89]	+1.73 [0.86, 2.55]
Sequence	FA	-5.15 [-6.12, -4.20]	-2.27 [-3.49, -1.18]
Sequence	TA + FA	-2.84 [-4.17, -1.62]	-0.53 [-1.81, 0.64]
Spatial patch	S-TA	+0.84 [0.07, 1.59]	+3.13 [2.25, 4.00]
Spatial patch	S-SA	+0.40 [-0.34, 1.18]	+0.88 [0.17, 1.65]
Spatial patch	S-TA + S-SA	+3.03 [2.08, 3.95]	+3.00 [2.20, 3.77]

Family	Variant	SWT-w7 gain [95% interval]	SW-w7 gain [95% interval]
Interleaved-month folds, four folds			
Sequence	TA	+0.03 [-1.89, 1.48]	+0.45 [-0.82, 1.65]
Sequence	FA	-1.55 [-2.58, -0.48]	-0.42 [-1.41, 0.62]
Sequence	TA + FA	-1.58 [-3.57, 0.20]	-2.33 [-4.21, -0.22]
Spatial patch	S-TA	-0.88 [-3.26, 1.28]	+0.53 [-1.78, 2.56]
Spatial patch	S-SA	+1.47 [0.13, 3.00]	+2.91 [1.58, 4.14]
Spatial patch	S-TA + S-SA	+0.88 [-1.27, 2.82]	+2.00 [0.41, 3.58]

Gain is Uniform RMSE minus attention RMSE in AQI points, so positive values favour attention; folds are weighted equally. Intervals are paired calendar-week block bootstraps. They cover test-week sampling only, not training-seed variation, and are unadjusted across the 36 comparisons.

A.7 Development-only event alerts

A separate 2025 historical test examined severe AQI (at least 300) and abrupt absolute daily change (at least 50 AQI points). Alert classifiers were fitted on October 2024 GRU predictions and thresholds chosen on November-December 2024, before the 2025 evaluation. Table 8 compares them with direct thresholds on the frozen forecast. Severe-event precision rose from 83.7% to 88.4%, but recall fell from 82.9% to 64.9%. For abrupt changes, the candidate produced only nine alerts and recall fell from 4.25% to 0.20%; its 88.9% precision is therefore not evidence of better detection. Neither classifier met the prespecified promotion check, and neither was deployed.

Table 8 Historical 2025 alert test; development candidates are not deployed.

Event	Method	Events	Alerts	Precision	Recall
AQI >= 300	Frozen forecast threshold	2,975	2,948	83.68%	82.92%
AQI >= 300	Candidate classifier	2,975	2,183	88.41%	64.87%
Change >= 50	Frozen forecast threshold	3,975	304	55.59%	4.25%
Change >= 50	Candidate classifier	3,975	9	88.89%	0.20%

All rows use the same 30,263 historical 2025 station-days. Precision and recall measure event alerts, not point-forecast accuracy.

B. Discussion

B.1 Why the average-error models differed

Persistence was a strong baseline in every forecast comparison, showing that recent AQI carries substantial predictive information. The much poorer 2024 environment-only Ridge score, alongside the better history-enhanced Ridge score, supports the importance of an AQI-history anchor in same-day estimation. The 28-day GRU improved on repeating yesterday's value in the 2025 cohort, consistent with useful information in a longer AQI sequence and known calendar/location inputs. That comparison does not isolate which individual GRU input caused the gain.

The 2026 weather-assisted XGBoost model predicts a correction to the previous-day AQI and uses archived forecasts available before the target day. Its lower paired error is consistent with a useful combination of that anchor, nonlinear residual modeling, and forecast-weather context. Because the comparator is a different architecture, this test alone cannot measure a weather-only causal contribution. The observed gain is a model-level result, not proof that any single weather variable explains it.

B.2 Where the gain weakened

The weather candidate flagged roughly three times as many sharp changes as GRU (799 versus 268), which helps explain its higher recall and lower precision. Its underprediction in high-AQI bands shows why a lower overall RMSE can coexist with weak performance on severe episodes. Rare extremes, incomplete station coverage, and different city weights make the pooled result insufficient as an operational safety criterion. Delhi contributed 12,387 of 30,263 rows in the 2025 GRU test (40.9%), so that pooled metric gives Delhi substantial influence.

The promotion rule required both sharp-change recall and precision to be at least as high as GRU's, in addition to positive paired RMSE gains. XGBoost missed the precision condition (52.82% versus 57.46%), so GRU remained the default next-day model despite XGBoost's lower average error. This decision follows the predeclared rule rather than choosing a model after viewing only its best metric.

B.3 Why attention accuracy is not explanation faithfulness

On SWT-w7, spatial attention improved the spatial-patch model on its own 2023 test, but FA and TA did not improve the sequence model. Table 7 shows that the S-SA and FA outcomes reverse on SW-w7, so they depend on the input set as well as the architecture. A learned weighting therefore cannot be assumed to help a given architecture. The faithfulness audit further separates accuracy from interpretability: attention rankings agreed less with replacement sensitivity than SHAP in all four comparisons. For S-SA under mean replacement, the highest-attention group's deletion changed output by 72.53 AQI points, close to 70.46 for a random ranking; SHAP's top group changed it by 73.76. This weak separation limits an explanation claim for that deletion test.

Replacement is an intervention on a fixed model, not a controlled experiment on atmospheric causes. Correlated inputs and unrealistic mean replacements can change rankings; the different same-city seasonal-donor correlations in Table 5 demonstrate that sensitivity. Thus, neither an attention heat map nor one perturbation score establishes the physical source of pollution.

B.4 Why cross-city conclusions differ from pooled conclusions

The corrected LOCO study directly tests transfer to an unseen city. It changes the previous inference that cross-city validation was still pending. The spatial S-TA+SA variant outperformed its within-family Uniform control on the equal-city average and in five cities, suggesting that joint temporal and spatial weighting can help this architecture transfer across locations. The gain is not uniform: three cities did not improve. The S-TA+SA gain also held on SW-w7 (Table 7), whereas sequence TA gained only on SW-w7. Sequence TA illustrates why aggregation matters: a small pooled RMSE gain occurred alongside a small city-macro RMSE loss. Delhi supplied 11,658 of the 29,923 LOCO station-days, so the pooled score is influenced strongly by that city's error. Equal-city averaging answers a different question from pooled station-day error.

The LOCO checkpoints were evaluated for predictive accuracy, not with the 300-row replacement-based faithfulness audit used for the separate chronological checkpoints. Therefore the transfer gain cannot be used to claim that their attention maps are faithful explanations. Likewise, the LOCO architecture uses different inputs and training folds from the operational GRU and weather XGBoost; its scores do not justify replacing the frozen next-day default.

B.5 Why higher alert precision did not justify promotion

The severe-event candidate issued 765 fewer 2025 alerts than the frozen forecast threshold. That more selective behavior increased precision but missed many additional true severe days, reducing recall by 18.05 percentage points. The abrupt-change candidate was still more selective: nine alerts for 3,975 observed events, with eight correct, so high precision describes a very small subset and almost no coverage. This is consistent with the difficulty of predicting sudden AQI changes from the current forecast and prior-day AQI alone. Both candidates failed the calibration promotion rule; their 2025 results are research findings only.

C. Inference

The research gap has three distinct answers. First, recent AQI is a strong baseline, while history-aware and weather-assisted models improved average error within their own paired historical cohorts. Second, the corrected LOCO experiment supplies evidence of partial cross-city transfer: S-TA+SA improved the eight-city average relative to S-Uniform on both input sets, but not every city; sequence TA improved the equal-city average only on SW-w7. Several other attention gains reversed between input sets (Table 7), so the evidence does not support a general claim that learned attention improves accuracy. Third, attention-based accuracy and explanation quality remain separate. On sampled chronological checkpoints, attention rankings agreed less with fixed-model replacement sensitivity than SHAP; the LOCO checkpoints have not yet received that audit. Neither analysis supports a causal claim about pollution sources.

Lower RMSE alone is insufficient for deployment. The weather XGBoost candidate failed the predeclared sharp-change precision condition, and the event classifiers lost too much recall and failed their own promotion checks. The frozen GRU therefore remains the next-day default. The 30% GRU / 70% XGBoost blend was selected on earlier development data but has no independent prospective score and is excluded from the validated comparisons.

The October-December 2026 evaluation period is still in progress. As of 2 November 2026, CPCB October exports existed for 104 of 107 stations, with 1,559 usable station-days inside the test window, and there were no immutable before-target forecast records, so there is no prospective metric to report. Those records cannot be produced from monthly CPCB exports: a record needs the previous day's AQI before target-day UTC midnight, and the portal offered no row for the current month. The frozen blend can still be scored after 31 December from later exports and archived forecasts; that would be an out-of-time test of a frozen model, not a record of live-issued forecasts. Historical and cross-city results should be interpreted as evidence for the specified evaluation settings, not as verified future performance.

Reproducibility note: Numerical results are traceable to the project's saved forecast metrics (results/forecast_2025 and results/forecast_2026), corrected LOCO summary (results/city_transfer/model_summary.csv), attention comparison across input sets (results/attention_stability/gains.csv), alert metrics (results/event_alerts/metrics.csv), attention audits (results/followthrough/oct12-v1), and prospective status (results/prospective/status.json).